

Appendix A**Dual-Channel Redundant Safety Control Architecture**

Architecture update for SAR / SHA documentation
Drone Remote Weapon Station (Human-in-the-Loop)

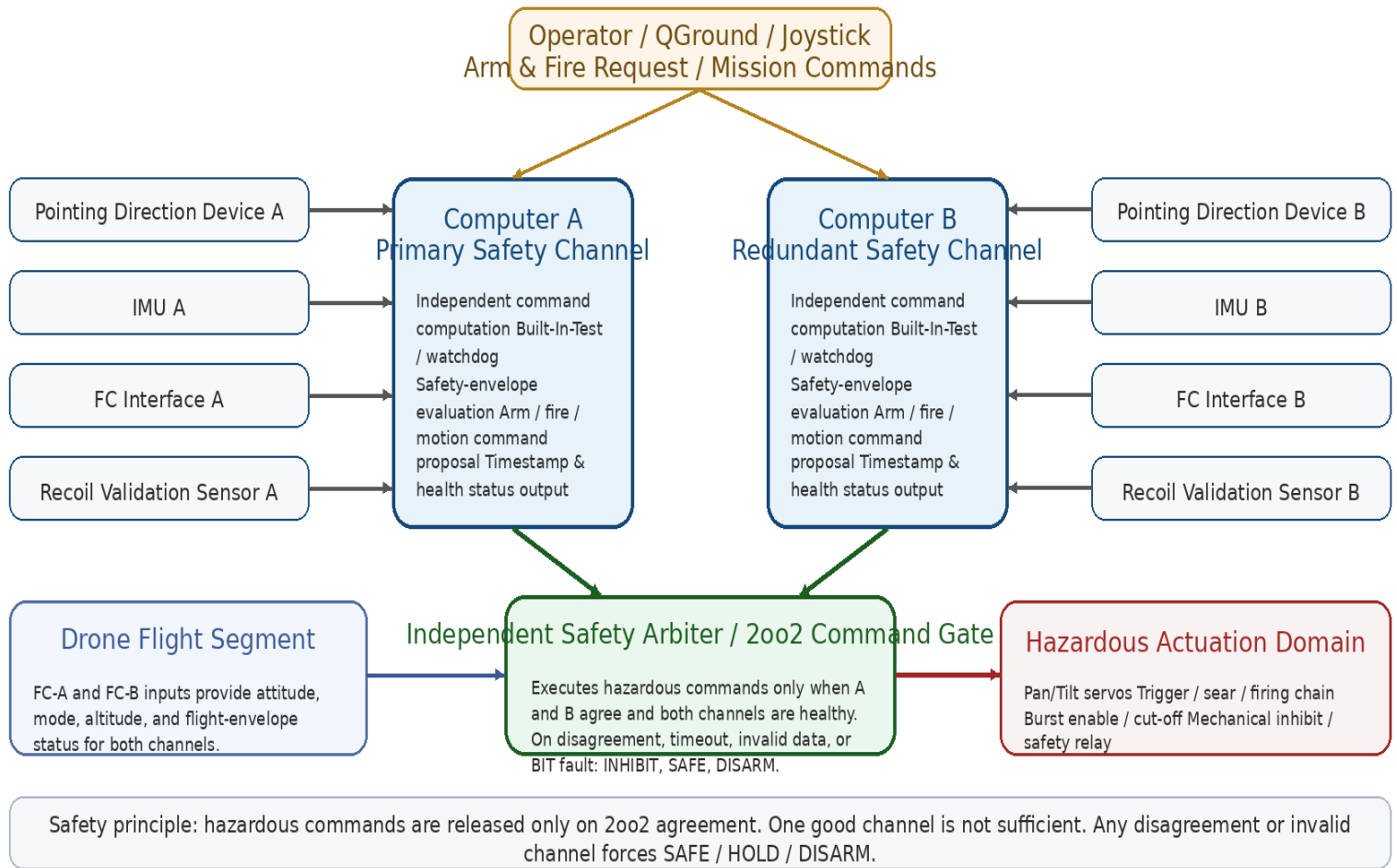
Safety upgrade

Replace the single hazardous command path with two independent control channels (Computer A and Computer B) and a separate 2oo2 command gate that releases hazardous commands only when both channels agree and are healthy.

1. Purpose and safety intent

- This appendix updates the architecture from a single-computer Jetson implementation to dual-channel redundant safety-control architecture. The new design is intended for hazardous-command release functions such as ARM, FIRE, burst enable, trigger actuation, and high-authority pan/tilt motion.
- The governing safety principle is 2oo2 command release: one valid channel is not sufficient. A hazardous command is executed only when Computer A and Computer B independently compute the same permitted action within defined tolerance and both channels remain healthy, current, and inside the safe operating envelope.
- **Any disagreement, stale data, timeout, failed built-in-test, watchdog event, invalid sensor input, or command-gate fault shall inhibit execution and force SAFE / HOLD / DISARM behavior.**

2. Proposed upgraded architecture diagram



3. Channel allocation and sensor independence

- Each control channel shall receive its own safety-critical input. The intent is independent computation, independent data validation, and independent health reporting before command release.

Item	Channel A	Channel B	Safety note
Computing node	Computer A	Computer B	Independent processing; no single computer may directly release a hazardous command.
Pointing direction	Pointing direction device A	Pointing direction device B	Use separate devices or separate independent acquisition chains.
Attitude / inertial	IMU A	IMU B	Cross-channel tolerance and plausibility checks required.
Flight status input	FC interface A	FC interface B	Independent channel interfaces to flight status; use separate buses/ports where practical.
Recoil Management Operation validation	Recoil sensor A	Recoil sensor B	Prefer diverse sensing principles if feasible.
Command decision	Command proposal A	Command proposal B	Only matching commands may pass to the arbiter.
Final release	—	—	Independent arbiter / safety gate performs 2oo2 comparison and releases or inhibits.

4. Safety-functional requirements

- SR-A1. Hazardous commands shall be generated independently by Computer A and Computer B using separate validated input sets.
- SR-A2. Hazardous command execution shall require 2oo2 agreement at an independent command gate; one correct computer shall not be sufficient for execution.
- SR-A3. The command gate shall compare command type, sign, magnitude, mode, validity, timestamp, and channel health before release.
- SR-A4. Each control channel shall perform built-in tests, watchdog supervision, sensor plausibility checks, and stale-data detection.
- SR-A5. If either channel fails BIT, loses a required sensor, exceeds timing limits, or produces a non-matching result, the system shall inhibit hazardous actuation and transition to SAFE / HOLD / DISARM.
- SR-A6. ARM and FIRE authority shall remain human-in-the-loop and shall be accepted only when both channels report a valid safe envelope, and the operator authorization chain is complete.
- SR-A7. Recoil-validity inputs shall be checked by both channels before firing and during burst execution; loss of recoil-control validity shall immediately inhibit continued firing.
- SR-A8. The actuation path for trigger/firing-chain release shall include a hardware safety interlock or relay that defaults to safe on loss of power, watchdog event, or gate fault.
- SR-A9. The architecture shall identify and mitigate common-cause failures, including shared power, shared clocks, shared communication buses, shared software defects, and EMI affecting both channels.

5. Command release logic and failure behavior

Condition	Arbiter decision	Required system response
A healthy + B healthy + commands agree	Release permitted	Execute approved hazardous command.
A healthy + B healthy + commands disagree	Do not release	Latch disagreement fault; SAFE / HOLD / DISARM.
A healthy + B invalid or timed out	Do not release	No single-channel execution; maintain inhibit.
A invalid + B healthy	Do not release	No single-channel execution; maintain inhibit.
Any required sensor invalid on either channel	Do not release	Annunciate fault and inhibit until reset / maintenance action.
Loss of power or arbiter fault	Default safe	Open inhibit path and de-energize hazardous actuation.

6. Verification evidence in SAR / SHA

Design feature	Purpose	Recommended evidence	Document linkage
Dual computers + 2oo2 gate	Prevent single-channel hazardous release	Architecture review, interface control document, bench demonstration	SAR appendix / SHA mitigation
Pointing A/B and IMU A/B cross-checks	Detect bad orientation data	Sensor fault injection, tolerance test, HIL logs	Hazard controls / verification matrix
FC-A / FC-B safe-envelope validation	Prevent firing outside flight margin	Flight-envelope tests, comm-fault tests	CONOPS safety / recoil-flight coupling
Recoil A/B validation	Verify recoil-cancel and inhibit on loss of control	Burst tests, induced recoil-sensor fault tests	SHA mitigation / FTA evidence
Independent safety relay / default-safe output	Ensure safe state on power or gate failure	Power interruption test, watchdog reset test	Safety device evidence
Common-cause analysis	Address shared failures	CCF worksheet, EMI test, shared-resource review	SAR residual risk rationale